3D Object Detection with Mesh R-CNN

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1 Goal of the work

The goal of this project was to investigate the applicability limits of the Mesh R-CNN model by testing it on images of objects from different categories and evaluating the quality of the predicted 3D meshes.

The experiment focused on three cases:

- An object from a category that exists in the Pix3D dataset.
- An object from another Pix3D category.
- An object outside the Pix3D training categories.

This structure allowed us to compare how well Mesh R-CNN performs on familiar categories and how it behaves when it receives an object that does not belong to its training distribution.

2 Theoretical Background

Mesh R-CNN is a model for joint 2D object perception and 3D shape prediction. The authors describe it as a system that takes a real-world RGB image, detects object instances, and predicts a 3D triangle mesh for each detected object. Unlike ordinary 2D detectors, Mesh R-CNN does not stop at bounding boxes or segmentation masks; it also estimates the object's full 3D shape.

The model is based on Mask R-CNN. Mask R-CNN predicts a category label, bounding box, and segmentation mask for each detected object. Mesh R-CNN extends this architecture by adding a mesh prediction branch. This branch first predicts a coarse voxel representation and then converts it into a mesh, which is refined using graph convolution layers operating over the mesh vertices and edges.

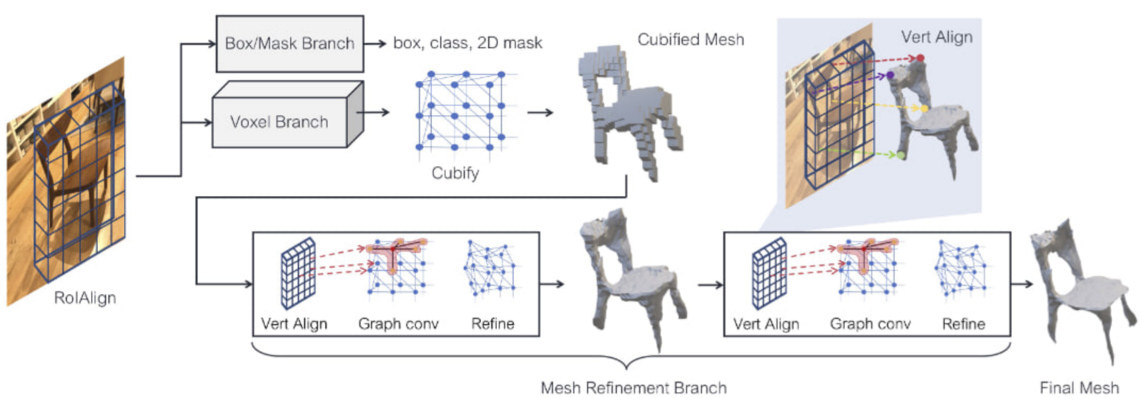

Figure 1: System overview of Mesh R-CNN. The paper augments Mask R-CNN with 3D shape inference. The voxel branch predicts a coarse shape for each detected object which is further deformed with a sequence of refinement stages in the mesh refinement branch.

The paper explains that this hybrid design is important because voxel predictions allow the model to represent different topologies, while mesh refinement improves the final surface geometry. This allows Mesh R-CNN to predict meshes with varying topology instead of being restricted to a fixed template. [1]

3 Methodology

The project was implemented in a Jupyter Notebook inside WSL. The following libraries and tools were used:

- Python 3.10
- PyTorch
- CUDA
- Detectron2
- PyTorch3D
- Mesh R-CNN

The pretrained Pix3D Mesh R-CNN model was used. The model was launched using the official demo script:

`demo/demo.py`

with the configuration file:

`configs/pix3d/meshrcnn_R50_FPN.yaml`

For the first two images, the `--onlyhighest` flag was used to keep only the highest-confidence prediction. For the out-of-distribution object, the model was allowed to generate multiple predictions so that its behavior on an unfamiliar object could be analyzed.

4 Selected Images

Figure 2: Chair, Sofa and Motorcycle

Figure 3: Chair, Sofa and Car

The **chair** and **sofa** were selected because they are close to the Pix3D furniture domain. The **motorcycle** and **car** was selected because it is not a Pix3D furniture category, making it useful for testing the model's generalization limits.

5 Results for the Chair Image

Figure 4: The predicted results for the chair image

For the chair image, the model predicted the class: **chair**, with a confidence score of: **1.000**

This is a correct semantic prediction. The chair belongs to a category that is close to the Pix3D training domain, so a high-confidence prediction was expected.

The generated mesh captured the general chair-like structure, including a large back/seat region and several supporting leg-like parts. However, the mesh quality was rough. The thin structures, especially the legs, were distorted and incomplete. This result is consistent with the Mesh R-CNN paper's explanation that the initial cubified mesh gives only a coarse shape and cannot accurately model fine structures such as chair legs without refinement [1].

Therefore, the model worked well for object recognition, but the 3D reconstruction quality was limited in detailed regions.

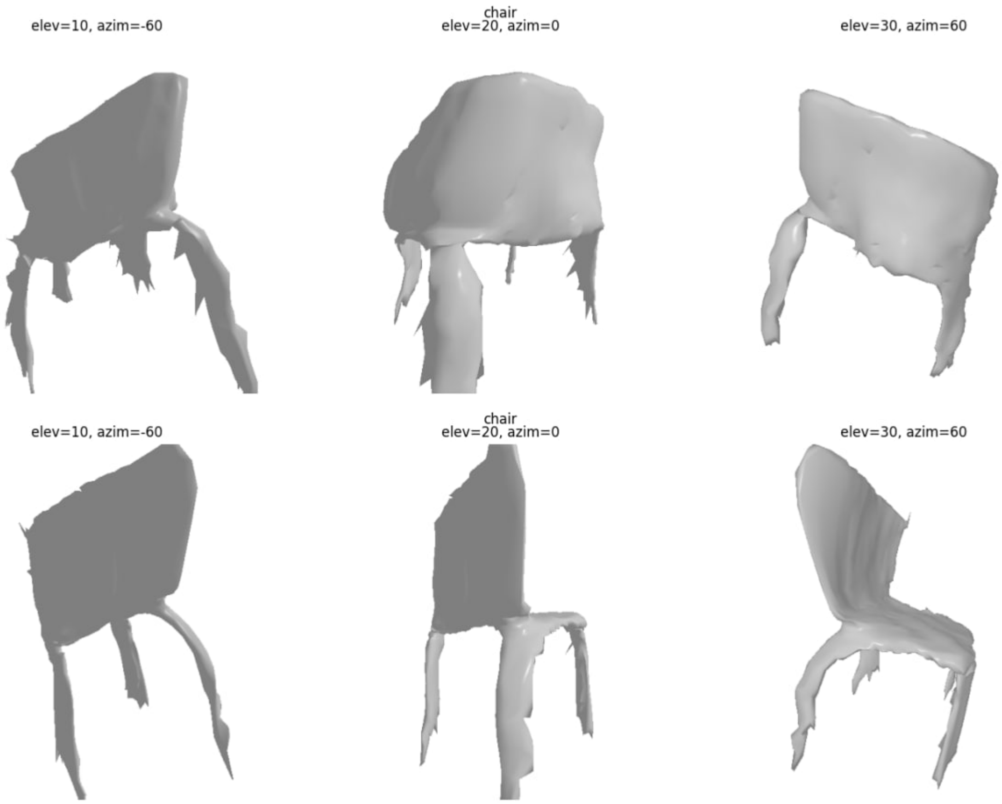

Figure 5: Rendered mesh visualization for Chair

This visualization step was important because the raw `.obj` file does not directly show the quality of the reconstructed shape. Rendering the mesh from multiple viewpoints made it possible to analyze whether the predicted 3D geometry preserved the main object structure, such as the chair backrest or possible object-specific details.

For the chair image, the rendered mesh showed a general chair-like shape, but the legs and thin supporting parts were distorted

Therefore, the rendered mesh views confirmed that Mesh R-CNN can recover approximate 3D structure for familiar Pix3D-like categories, but it struggles with fine details and out-of-distribution objects.

6 Results for the Sofa Image

Figure 6: The predicted results for the sofa image

For the sofa image, the model predicted the class: **sofa**, with a confidence score of: **1.000**

This was also a correct prediction. The sofa belongs to another known furniture category, so the model successfully recognized it.

The predicted mesh represented the general elongated volume of the sofa, but the reconstruction was still simplified. The global shape was visible, but details such as cushions, armrests, soft surfaces, and small legs were not accurately reconstructed.

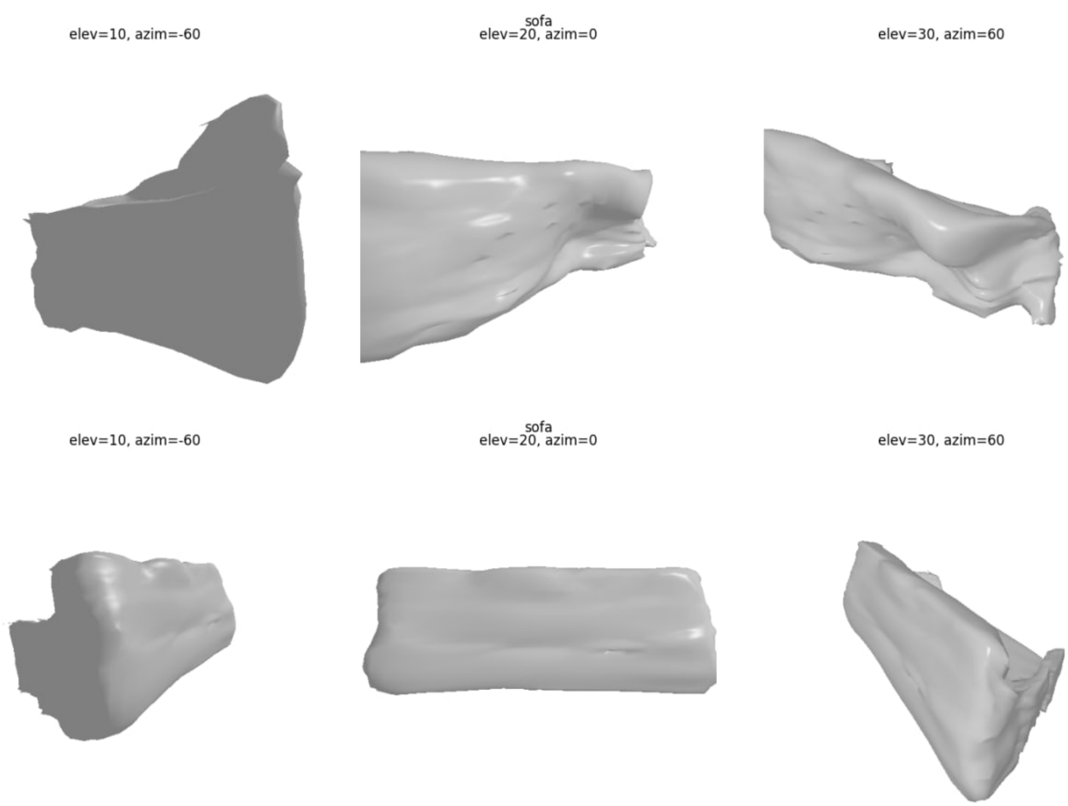

Figure 7: Rendered mesh visualization for Sofa

This reflects one of the main challenges of single-image 3D reconstruction: the model must infer a full 3D shape from a single RGB image. Mesh R-CNN is designed to address this by combining 2D recognition with 3D shape inference, but the predicted mesh is still an approximation of the real object geometry. [1]

7 Results for the Motorcycle and Car Image

Figure 8: The predicted results for the motorcycle image

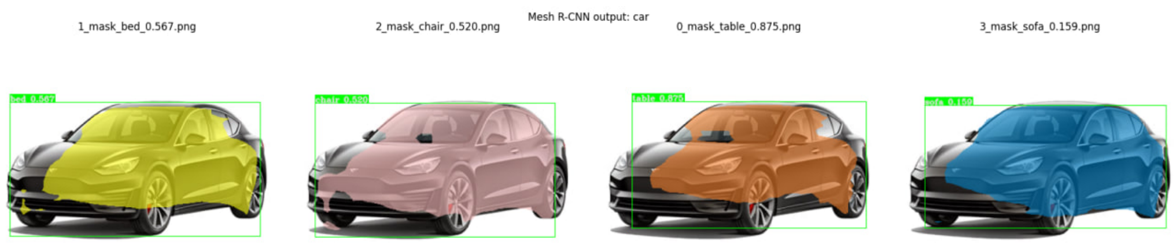

Figure 9: The predicted results for the car image

The motorcycle image was used as an out-of-distribution example because it does not belong to the Pix3D furniture categories.

For this image, the highest-confidence prediction was: **chair**, with a confidence score of: **0.989**

The model also produced other predictions, including:

Table 1: Top Mesh R-CNN predictions for the motorcycle image

Predicted class	Confidence score	Mesh file
chair	0.989	`0_mesh_chair_0.989.obj`
bed	0.749	`1_mesh_bed_0.749.obj`
table	0.724	`2_mesh_table_0.724.obj`
table	0.589	`3_mesh_table_0.589.obj`
sofa	0.436	`4_mesh_sofa_0.436.obj`
tool	0.415	`5_mesh_tool_0.415.obj`
tool	0.336	`6_mesh_tool_0.336.obj`
tool	0.305	`7_mesh_tool_0.305.obj`
chair	0.280	`8_mesh_chair_0.280.obj`
sofa	0.204	`9_mesh_sofa_0.204.obj`

This result is important because the model assigned a very high confidence score to an incorrect category. Instead of identifying the motorcycle as unknown, the model forced it into one of the known Pix3D categories.

The predicted mesh did not resemble a motorcycle. It did not reconstruct wheels, handlebars, the frame, or other motorcycle-specific geometry. Instead, the generated mesh looked like a distorted furniture-like object. This demonstrates that Mesh R-CNN does not have reliable out-of-distribution awareness: if an object is outside the training categories, the model may still produce a confident but incorrect prediction.

This limitation is understandable because the Mesh R-CNN paper focuses on known object categories in datasets such as ShapeNet and Pix3D. In the Pix3D setting, the task is to jointly detect and predict 3D shapes for known object categories. [1]

Table 2: Top Mesh R-CNN predictions for the car image

Predicted class	**Confidence score**	**Mesh file**
table	0.875	`0_mesh_table_0.875.obj`
bed	0.567	`1_mesh_bed_0.567.obj`
chair	0.520	`2_mesh_chair_0.520.obj`
sofa	0.159	`3_mesh_sofa_0.159.obj`

8 Conclusion

In this project, Mesh R-CNN was tested on three images: a chair, a sofa and a motorcycle.

The model correctly recognized the chair and sofa with confidence scores of 1.000. This shows that Mesh R-CNN works well for object categories similar to the Pix3D training data. However, the predicted meshes were rough and simplified, especially around thin or detailed parts.

For the motorcycle image, the model failed to recognize the real object category and predicted it as a chair with confidence 0.989. This demonstrates that Mesh R-CNN does not generalize reliably to completely new object types outside its training categories.

Overall, Mesh R-CNN is useful for approximate 3D reconstruction of familiar object categories. Its main strength is combining 2D object detection with 3D mesh prediction in a single model. Its main limitation is that mesh quality depends strongly on object category, image quality, and similarity to the training distribution.

References

[1] G. Gkioxari, J. Malik, and J. Johnson, "Mesh r-cnn," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9785–9795, 2019.